

Gauri Godghase

Data Scientist

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WORK EXPERIENCE

San Jose State University, CA | Data Science Researcher ([LLM implementation paper](#))

January 2024 – Present

- Processed and structured large-scale unstructured datasets using **Python (Pandas)**, applying embedding models like **BERT, Word2Vec, and GloVe** to extract semantic meaning and prepare data for downstream **LLM tasks**.
- Developed and iterated on NLP pipelines for classifying human vs. LLM-generated text over datasets exceeding 1.5M samples, applying models built in **PyTorch** and **scikit-learn** to handle diverse linguistic patterns and edge cases.
- **Fine-tuned transformer-based LLMs** for classification and generation tasks, experimenting with different architectures and training configurations to improve model accuracy and generalization on edge cases.
- Utilized **SQL and MongoDB** for querying and managing large datasets during model development, ensuring efficient access to relevant data subsets for training and evaluation.
- Built an interactive **front-end tool** using **JavaScript** to visualize model results and support exploratory analysis, linking insights from backend processing to a more intuitive user-facing interface.

Workiva, CA | Data Scientist

May 2023 – December 2023

- Designed generative AI powered chatbot using **LLMs (e.g., GPT, LLaMA)** to handle domain-specific tasks such as question answering, summarization, and content drafting based on unstructured input data.
- Built **retrieval-augmented generation (RAG)** pipelines in an **AWS-based environment**, integrating OpenSearch with vector databases to enable high-precision retrieval for context-aware generation.
- Benchmarked multiple **LLM architectures** through controlled experiments, identifying the most effective models based on performance, latency, and cost considerations within cloud constraints.
- Leveraged frameworks such as **LangChain and Hugging Face Transformers** to build reusable, modular components—enabling experimentation with memory, agent workflows, and dynamic tool usage.
- Explored **multi-agent orchestration** methods to coordinate interactions between LLMs and downstream systems, enabling more structured decision-making and complex task execution.
- Used **AWS services (e.g., S3, Lambda, EC2)** to manage datasets, serve model components, and support RAG pipeline workflows at scale.
- Utilized Metaflow and MLflow to manage machine learning workflows and track experiments end-to-end — from data preprocessing and model training to parameter tuning and versioned artifact logging — ensuring reproducibility and streamlined collaboration across teams.

Cognizant, India | Data Scientist

March 2020 – July 2022

- Designed and implemented **scalable ML pipelines in Python** for demand forecasting, integrating batch ingestion, preprocessing, model training, and evaluation components to reduce inventory mismatches and improve operational efficiency.
- Engineered **robust ETL workflows** to automate ingestion and transformation of multi-format data (CSV, APIs, SQL) into training-ready datasets, ensuring high data quality and schema consistency across environments.
- Built and tuned **supervised learning models (Random Forest, XGBoost, Logistic Regression)** using scikit-learn for fraud detection and churn prediction, leading to a 20% lift in precision through feature optimization and cross-validation.

TECHNICAL SKILLS

Programming & Scripting: Python (Pandas, NumPy, Matplotlib, SciPy, Scikit-learn, PyTorch, TensorFlow), SQL (MySQL, PostgreSQL), JavaScript (Node.js, basic front-end)

Machine Learning & AI: LLM Fine-Tuning (GPT, LLaMA), Retrieval-Augmented Generation (RAG), Prompt Engineering, Supervised & Unsupervised Learning, Deep Learning (CNNs, RNNs), NLP (BERT, Word2Vec, spaCy), Model Evaluation (Cross-validation, SHAP, ANOVA)

Data Engineering & Pipelines: Data Ingestion & Structuring, ETL Development, Workflow Orchestration (Apache Airflow, AWS Step Functions), Experiment Tracking (MLflow, Metaflow)

Cloud & DevOps: AWS (S3, Lambda, EC2, SageMaker, Redshift), Docker, Git/GitHub, Vector Stores (FAISS, Pinecone), OpenSearch

Databases & Storage: SQL (MySQL, PostgreSQL), NoSQL (MongoDB), Data Warehousing (Redshift, BigQuery)

Visualization & Reporting: Power BI, Tableau, Plotly, Dash; building lightweight internal tools with JS for data interaction

Tools & Frameworks: LangChain, Hugging Face Transformers, Jupyter Notebooks, VSCode, Linux CLI

Soft Skills & Domain Exposure: Rapid Prototyping, Cross-functional Collaboration, A/B Testing, Stakeholder Communication, Experience in Agile and startup-like environments

EDUCATION

Masters in Computer Science | San Jose University, CA, USA.

Bachelors in Computer Engineering | Maharashtra Institute of Technology, Pune, India.